MCA FULL STACK DEVELOPMENT

Practical Report — Question 1

Basic HTML Document Structure & Text Formatting

Course	MCA IBM — 1st Semester
Subject	Full Stack Development (CAPCMC105T)
Practical file	intro.html
Stylesheet	style.css

1. Objective

To create a single HTML page demonstrating the basic HTML document structure and text-formatting elements, including headings, paragraphs, bold and italic text, small text, superscript, subscript, line breaks, preformatted text, span, strong, code, and an HTML comment.

2. Software and Files Required

• Visual Studio Code
• A web browser such as Google Chrome
• Live Server extension (for local preview)
• intro.html and style.css in the same practical folder

3. HTML Concepts Explained

Document declaration and root: <!DOCTYPE html> declares HTML5; <html lang="en"> is the root element and identifies the document language.

Head section: <head> contains metadata, the browser-tab title, and the link to the stylesheet. The meta charset declaration supports UTF-8 text; the viewport declaration helps responsive display.

Body section: <body> contains the content rendered in the browser. <header>, <main>, <section> and <footer> organize the page into meaningful regions.

Headings and paragraphs: <h1> through <h3> demonstrate heading levels. <p> creates paragraphs.

Formatting: <b> presents bold text, <i> italic text, <small> smaller text, <sup> superscript, and <sub> subscript. indicates importance.

Spacing and inline elements:
 inserts a line break; <pre> preserves spaces and line breaks; groups inline text for styling; <code> marks code-related text. HTML comments use <!-- comment --> and are not displayed as page content.

4. CSS Concepts Explained

Selector: body, main, pre and code select HTML elements; .page-header, .content-card and .highlight select elements by class.

Properties: font-family selects typefaces; background-color and color control colors; margin controls outside spacing; padding controls inside spacing; border draws an outline; border-radius rounds corners; max-width limits content width; text-align aligns text; line-height controls line spacing.

Responsive styling: @media (max-width: 600px) applies reduced spacing on narrower screens.

5. Complete Source Code — intro.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>HTML Basics and Text Formatting</title>
    <!-- Connect the external CSS file -->
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <!-- Main heading -->
    <header class="page-header">
        <h1>Introduction to HTML</h1>
        <p>A beginner's guide to HTML text formatting.</p>
    </header>

    <main>
        <!-- Demonstration of heading levels -->
        <section class="content-card">
            <h2>1. Heading Levels</h2>
            <h1>Heading Level 1</h1>
            <h2>Heading Level 2</h2>
            <h3>Heading Level 3</h3>
        </section>

        <!-- Demonstration of paragraphs -->
        <section class="content-card">
            <h2>2. Paragraphs</h2>
            <p>HTML stands for <strong>HyperText Markup Language</strong>.
               It defines the structure of a webpage.</p>
            <p>HTML can display <b>bold text</b>, <i>italic text</i>,
               and <small>smaller text</small>.</p>
            <p>HTML is used to create web pages. Learning its elements is
               an important first step in web development.</p>
        </section>

        <!-- Demonstration of superscript and subscript -->
        <section class="content-card">
            <h2>3. Superscript and Subscript</h2>
            <p>Mathematics: x<sup>2</sup> + y<sup>2</sup></p>
            <p>Chemistry: H<sub>2</sub>O</p>
        </section>

        <!-- Demonstration of line breaks -->
        <section class="content-card">
            <h2>4. Line Break</h2>
            <p>MCA Student<br>
               Full Stack Development<br>
               HTML Practical - Question 1</p>
        </section>

        <!-- Demonstration of preformatted text -->
        <section class="content-card">
            <h2>5. Preformatted Text</h2>
            <pre>
Name:  Student
Course: MCA
Subject: Full Stack Development
            </pre>
        </section>

        <!-- Demonstration of span and code -->
        <section class="content-card">
            <h2>6. Span, Strong and Code</h2>
            <p>This is a <span class="highlight">highlighted phrase</span>.</p>
            <p><strong>Important:</strong> Save your HTML file before testing it.</p>
            <p>Example code: <code>&lt;h1&gt;Hello World&lt;/h1&gt;</code></p>
        </section>
    </main>

    <!-- Footer -->
    <footer>
        <p>HTML Practical — Question 1</p>
    </footer>
</body>
</html>
```

6. Complete Source Code — style.css

```
/* Basic page styling */
body {
    font-family: Arial, sans-serif;
    background-color: #f1f5f9;
    color: #1e293b;
    margin: 0;
    padding: 20px;
    line-height: 1.6;
}

/* Main page header */
.page-header {
    background-color: #1e3a5f;
    color: white;
    text-align: center;
    padding: 25px;
    border-radius: 8px;
}

/* Main content container */
main {
    max-width: 900px;
    margin: 20px auto;
}

/* Individual content sections */
.content-card {
    background-color: white;
    margin-bottom: 18px;
    padding: 20px;
    border: 1px solid #cbd5e1;
    border-radius: 8px;
}

/* Highlighted inline text */
.highlight {
    color: #075985;
    background-color: #e0f2fe;
    padding: 3px 6px;
}

/* Preformatted text block */
pre {
    background-color: #f8fafc;
    border: 1px solid #cbd5e1;
    padding: 12px;
    overflow-x: auto;
}

/* Inline code */
code {
    background-color: #e2e8f0;
    padding: 3px 6px;
    border-radius: 4px;
    font-family: Consolas, monospace;
}

/* Footer */
footer {
    text-align: center;
    margin-top: 25px;
    color: #475569;
}

/* Improve readability on smaller screens */
@media (max-width: 600px) {
    body {
        padding: 10px;
    }

    .content-card {
        padding: 14px;
    }
}
```

7. Output Screenshot

INSERT ACTUAL BROWSER SCREENSHOT HERE

Open intro.html with Live Server in Chrome, capture the rendered webpage, and insert the screenshot before submitting this report.

The screenshot has not been fabricated or captured from your computer.

8. Verification Checklist

- ■ Document structure: DOCTYPE, html, head, title and body
- ■ Three heading levels: h1, h2 and h3
- ■ Two or three paragraphs
- ■ b, i, small, sup and sub
- ■ br and pre
- ■ span, strong and code
- ■ Meaningful title and at least one HTML comment
- ■ CSS loads correctly and output checked in a browser

9. Result

The practical is ready for final verification. After confirming the webpage in the browser and inserting the actual output screenshot, the report can be used as the individual Q1 submission PDF.

Submission note: This is a report draft. Replace the screenshot placeholder with a genuine browser screenshot and confirm the source code matches the files you ran before submitting.